On the Additive Composition of Two and Two

A proof by categorical linearization, derived stabilization, Hodge theory, and heat-kernel descent

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Abstract

We determine the value of the expression $2 + 2$ in the ordinary natural numbers. Rather than evaluate it at the level at which it is posed, we transport the problem through the category of finite sets, the Grothendieck group of finite-dimensional vector spaces, a stabilized derived category, and the de Rham complex of a disconnected compact Riemannian manifold. The resulting elliptic operator is shown to have a heat supertrace independent of time, metric, and several decisions that did not need to be made. Positive eigenvalues cancel in pairs, leaving a finite obstruction to further abstraction. An explicit identification of the surviving harmonic modes with a standard four-element set yields the desired integer. We provide a curvature check, a spectral sequence, a determinant calculation, and a discussion of why none of these additions improves the result. The proof is unconditional, except for its reliance on established mathematics.

Keywords. Addition; finite cardinality; unnecessary generalization; elliptic operators; resistance to simplification.

Scope. This is mathematical satire built from standard mathematics. No novelty is claimed for the conclusion. The expenditure of machinery is the contribution.

Reader's advisory

The reader is assumed to know what two is. This assumption is revisited in Appendix A, after it has supported most of the paper. The impatient reader may proceed directly to the conclusion. The patient reader is asked to reconsider.

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1 Introduction and statement of the problem

The additive composition of small natural numbers is commonly regarded as settled. In the particular case considered here, this confidence often precedes the specification of a category, a coefficient field, or even a Riemannian metric. The present paper addresses the absence of these structures, although not their necessity.

Our starting point is the expression

$$2 + 2. \tag{1.1}$$

One could apply the recursive definition of addition immediately. Doing so would terminate the investigation before a single compact operator had appeared. We instead seek a route on which the answer remains unchanged while its surroundings become progressively more difficult to recognize.

Theorem 1.1 (Main theorem)**.** *In the natural numbers with their usual addition,*

$$2 + 2 = 4. \tag{1.2}$$

The numerical specificity of the theorem is intentional. We do not attempt a systematic treatment of $2 + 3$, whose inclusion would require changes to several labels. Nor do we claim that the method is optimal. There is no maximum length of a proof: one may insert arbitrarily many valid lemmas, identity maps, or contractible complexes. Our more modest aim is to make the overhead substantial, explicit, and mathematically accountable.

The strategy is to manufacture an invariant that remembers the input and then insist on recovering it by analysis. Each copy of two will be represented by the even cohomology of a sphere. Their union will be analyzed with the Hodge–de Rham operator. The nonzero spectrum will be dismissed only after we have paid for its existence. The remaining harmonic modes will be placed in correspondence with four named coordinates. Finally, a rank map will return the computation to the integers, where it began.

1.1 What kind of proof is being offered

There is an elementary derivation of the theorem from the recursion axioms of arithmetic. It appears in Appendix A. The main text is a deliberately elaborate derivation *inside* ordinary mathematics. It is not a claim that spectral theory can be developed without the arithmetic it is later used to certify.

This distinction matters. The usual statements of dimension theory, finite direct sums, and heat-trace formulas already use natural-number arithmetic. Hiding that fact under a large integral sign would make the paper incorrect rather than excessive. We therefore disclose the dependency and proceed with the excessive option.

Our constructions give mutually compatible certificates for the same elementary identity. They are not logically independent foundations of arithmetic. Independence would be an additional result, and the current manuscript has already exceeded the mathematical needs of its principal one.

1.2 Conventions and admissible shortcuts

All sets used to model the input are finite. All vector spaces are finite-dimensional unless they consist of differential forms or their Hilbert-space completions. All manifolds are smooth, compact, oriented, and without boundary. The Laplacian has nonnegative spectrum. The symbol + in the theorem denotes ordinary addition, not concatenation, exclusive-or, or an operation selected after observing the answer.

We use $\cong$ for isomorphism and $=$ for equality. Isomorphic vector spaces need not be identical sets, and equality in a Grothendieck group does not generally imply isomorphism of the original objects. These distinctions will be respected even when they delay the arrival of a very small integer.

Convention 1.2 (Premature evaluation)**.** An observation that already establishes the theorem may be recorded without being allowed to end the paper.

This is a convention about exposition, not a new rule of inference. The mathematics would be entitled to stop much earlier than we do.

2 Finite sets and the management of distinctness

Let

$$A = \{a_0, a_1\}, \qquad B = \{b_0, b_1\}, \tag{2.1}$$

where the displayed symbols are distinct. To avoid relying on this typographical arrangement, form the tagged disjoint union

$$U = (A \times \{L\}) \cup (B \times \{R\}), \qquad L \neq R. \tag{2.2}$$

The tags are retained even though the original sets were already disjoint. This provides redundancy at a stage where failure was unlikely.

Write $\iota_A(a) = (a, L)$ and $\iota_B(b) = (b, R)$. Equality of ordered pairs implies that each map is injective. If $\iota_A(a) = \iota_B(b)$, equality of second coordinates would imply $L = R$, contrary to construction. Thus no element supplied by one input can silently satisfy the administrative requirements of the other.

Lemma 2.1 (Separation of contributions)**.** *The images of ι_A and ι_B are disjoint and exhaust U.*

Proof. Disjointness was established by the tags. Exhaustion follows from the definition of the union in (2.2). There is no third clause in that definition in which an additional element could reside. □

Finite-cardinal addition is represented by disjoint union:

$$|A| + |B| = |A \sqcup B|. \tag{2.3}$$

Consequently, determining the original expression amounts to identifying the finite cardinality of U. The objects are already written down. To make the identification less immediate, we now replace them by a universal property.

2.1 The comparison object

Let S denote the successor operation and set

$$\underline{4} = \{0, S(0), S^2(0), S^3(0)\}. \tag{2.4}$$

This is the von Neumann ordinal $S^4(0)$ when successors are realized set-theoretically. We also use a relabeled copy

$$Q = \{q_0, q_1, q_2, q_3\}. \tag{2.5}$$

The map $q_j \mapsto S^j(0)$ identifies these two sets. Calling Q a four-element set therefore refers to a specified standard cardinal, not a number inferred from the eventual heat equation.

One may already match the displayed members of U and Q. We defer the matching table to Appendix B, where it can be presented as the output of the preceding infrastructure.

3 A categorical reformulation of putting things together

Work in the category FinSet of finite sets and all functions. For any finite set T, restriction to the two tagged components gives a function

$$\mathrm{Hom}(U, T) \longrightarrow \mathrm{Hom}(A, T) \times \mathrm{Hom}(B, T), \qquad h \longmapsto (h\iota_A, h\iota_B). \tag{3.1}$$

Given a pair (f, g) on the right, define

$$h(a, L) = f(a), \qquad h(b, R) = g(b). \tag{3.2}$$

Lemma 2.1 makes these prescriptions compatible and exhaustive. The two constructions are inverse. Moreover, postcomposition by a map $T \to T'$ commutes with restriction. Thus (3.1) is natural in T.

Proposition 3.1 (Coproduct certification)**.** *The object U, equipped with ι_A and ι_B, is a coproduct of A and B in* FinSet.

Proof. The preceding natural bijection is the coproduct universal property. □

This replaces the statement that the elements are all present by the statement that every possible function out of them is present in the correct way. The second statement is more expensive to read and contains the first as a special case.

3.1 Uniqueness and coherence

Suppose U' is another coproduct with structure maps j_A, j_B. The universal property supplies unique maps $u : U \to U'$ and $v : U' \to U$ preserving the input maps. Both vu and id_U have the same restrictions to A and B, so uniqueness gives $vu = \mathrm{id}_U$. Similarly $uv = \mathrm{id}_{U'}$.

Thus the choice of a concrete implementation of disjoint union does not affect the isomorphism class. We have proved this rather than allowing the word “the” in “the disjoint union” to bear the entire load.

Associativity is handled by the map that removes and reinstates tags:

$$(X \sqcup Y) \sqcup Z \cong X \sqcup (Y \sqcup Z). \tag{3.3}$$

The two composite maps around the usual associativity pentagon agree on every tagged input, hence agree everywhere. Our original expression has only two summands and does not need this pentagon. Its absence would nevertheless have left the larger program underdeveloped.

Remark 3.2. The universal property proves that U is determined up to a unique isomorphism preserving its input maps. It does not say that the underlying set has only one automorphism. Forgetting the input maps restores permutations. Keeping track of this distinction is one of the few genuine services performed by the present section.

4 Linearization and passage to a Grothendieck group

For a finite set X, let

$$F(X) = \mathbb{R}[X] = \bigoplus_{x \in X} \mathbb{R} e_x \tag{4.1}$$

be the vector space freely generated by its elements. A function $f : X \to Y$ induces $F(f)(e_x) = e_{f(x)}$. This respects identities and composition, so F is a functor from FinSet to $\mathsf{Vect}_{\mathbb{R}}^{\mathrm{fd}}$.

There is a canonical isomorphism

$$F(A \sqcup B) \cong F(A) \oplus F(B), \tag{4.2}$$

obtained by sending a tagged basis vector to the corresponding summand. Its inverse inserts the tag. We have changed a finite collection of symbols into an uncountable collection of linear combinations, without acquiring any additional information relevant to the question.

4.1 Why a group is required

The isomorphism classes of finite-dimensional real vector spaces form a commutative monoid under direct sum. We replace this monoid by its Grothendieck group $K_0(\mathbb{R})$. Concretely, this is generated by symbols $[V]$ with relations

$$[V] = [V'] + [V''] \tag{4.3}$$

for short exact sequences $0 \to V' \to V \to V'' \to 0$. Over a field every such sequence splits. The standard group-completion construction and its relation to K_0 are described in [2, Chapter II].

The map

$$\rho : K_0(\mathbb{R}) \longrightarrow \mathbb{Z}, \qquad [V] \longmapsto \dim_{\mathbb{R}} V \tag{4.4}$$

is well defined by additivity of dimension. It is an isomorphism: every vector space has a finite basis, so its class is an integer multiple of $[\mathbb{R}]$, and $\rho([\mathbb{R}]) = 1$. In particular, ρ has no kernel in which an arithmetic discrepancy could disappear.

Proposition 4.1 (Linear certificate)**.** *In $K_0(\mathbb{R})$,*

$$[F(U)] = [F(A)] + [F(B)]. \tag{4.5}$$

Proof. Apply the defining relation to the split short exact sequence with middle term $F(A) \oplus F(B)$, and use (4.2). □

The introduction of additive inverses is not demanded by the input, which contains no subtraction. It will become useful once we create terms that need to cancel. This is a familiar pattern: a complication may justify the machinery required to remove it.

4.2 Characteristic and rank

The target of ρ is $\mathbb{Z}$, even if one chooses a coefficient field of positive characteristic. Dimension is an integer, not the image of that integer in the coefficient field. Thus replacing $\mathbb{R}$ by $\mathbb{F}_2$ would not turn the dimension of a nonzero vector space into zero merely because the scalar $1 + 1$ vanishes.

We retain $\mathbb{R}$ so that differential forms can enter without a change of coefficients. The use of real numbers here is a logistical decision, not a claim that four requires completeness.

5 Derived stabilization and the controlled introduction of nothing

A vector space can be regarded as a cochain complex concentrated in degree zero. This permits us to pass from $\mathsf{Vect}_{\mathbb{R}}^{\mathrm{fd}}$ to the category of bounded complexes and then to its derived category by inverting quasi-isomorphisms. In this setting, one may replace an object by a more elaborate resolution without changing its cohomology.

Let $r \geq 0$ and define a complex $E_r^\bullet$ by

$$E_r^1 = \mathbb{R}^r, \qquad E_r^2 = \mathbb{R}^r, \qquad d^1 = \mathrm{id}_{\mathbb{R}^r}, \tag{5.1}$$

with all other terms and differentials zero. This complex is acyclic. Indeed, its only nontrivial differential is both injective and surjective. More strongly, it is contractible: set $h^2 = \mathrm{id} : E_r^2 \to E_r^1$ and all other components of h to zero. Then

$$dh + hd = \mathrm{id}_{E_r^\bullet}. \tag{5.2}$$

Now form

$$\widetilde{F}(U)^\bullet = F(U)[0] \oplus E_r^\bullet. \tag{5.3}$$

Projection onto $F(U)[0]$ is a quasi-isomorphism. The auxiliary coordinates have entered in adjacent degrees and are paired by an identity map. They are therefore guaranteed to contribute nothing, although this guarantee becomes visible only after the construction has been described.

Lemma 5.1 (Euler-class invariance)**.** *For a bounded complex $C^\bullet$ of finite-dimensional vector spaces,*

$$\sum_j (-1)^j [C^j] = \sum_j (-1)^j [H^j(C^\bullet)] \quad \textit{in } K_0(\mathbb{R}). \tag{5.4}$$

Proof. Put $Z^j = \ker d^j$ and $B^j = \operatorname{im} d^{j-1}$. The short exact sequences

$$0 \longrightarrow Z^j \longrightarrow C^j \longrightarrow B^{j+1} \longrightarrow 0, \tag{5.5}$$

$$0 \longrightarrow B^j \longrightarrow Z^j \longrightarrow H^j \longrightarrow 0 \tag{5.6}$$

give $[C^j] = [B^j] + [H^j] + [B^{j+1}]$. Multiply by $(-1)^j$ and sum. The boundary classes cancel in adjacent degrees. Boundedness ensures that no convergence question has been smuggled into this finite cancellation. □

In particular, the Euler class of (5.3) is

$$[F(U)] - [\mathbb{R}^r] + [\mathbb{R}^r] = [F(U)]. \tag{5.7}$$

The parameter r is completely arbitrary. A reader who considers the proof insufficiently long may increase it, although merely writing r larger will not increase the number of pages. For that, one must display the coordinates.

5.1 A grading warning

We will shortly replace $F(U)[0]$ by a geometric complex whose cohomology occurs in degrees zero and two. These complexes are *not* quasi-isomorphic as $\mathbb{Z}$-graded complexes. Their Euler classes and their parity-folded cohomology agree after the explicit identifications below. This is the precise comparison we need.

Confusing equality of Euler characteristics with quasi-isomorphism would save a sentence and lose the theorem's mathematical integrity. We have kept the sentence.

6 Geometric realization by a disconnected surface

Let S^2_A and S^2_B be two labeled copies of the unit sphere and define

$$M = S^2_A \sqcup S^2_B. \tag{6.1}$$

The disjoint union is again compact and has no boundary. Each component receives its standard orientation and round metric. The labels are retained so that the original two contributions remain traceable after the problem has become a differential equation.

The cellular cochain complex of a sphere has a generator in degree zero, a generator in degree two, and no other generators. This comes from the CW structure obtained by attaching a two-cell to a point by the constant attaching map. Since the degree-one group vanishes, both differentials are zero. Cellular and singular cohomology agree; the standard framework is developed in [1].

Consequently,

$$H^j(S^2;\mathbb{R}) \cong \begin{cases} \mathbb{R}, & j = 0, 2, \\ 0, & \text{otherwise.} \end{cases} \tag{6.2}$$

Each sphere therefore provides one cohomology generator for each member of one of our input pairs.

6.1 An alternative calculation, in case the cell was too easy

Cover S^2 by $V = S^2 \setminus \{\text{north pole}\}$ and $W = S^2 \setminus \{\text{south pole}\}$. Both are contractible, while $V \cap W$ deformation retracts onto a circle. In the Mayer–Vietoris sequence, the degree-zero restriction map is

$$\mathbb{R} \oplus \mathbb{R} \longrightarrow \mathbb{R}, \qquad (x, y) \longmapsto x - y. \tag{6.3}$$

Its kernel is the diagonal and it is surjective. Exactness therefore gives $H^0(S^2; \mathbb{R}) \cong \mathbb{R}$ and $H^1(S^2; \mathbb{R}) = 0$. The next nontrivial portion identifies $H^2(S^2; \mathbb{R})$ with $H^1(S^1; \mathbb{R}) \cong \mathbb{R}$. This recovers (6.2) by cutting the sphere into two pieces and repairing the cut with an exact sequence.

The open cover is not needed later. It is included to ensure that the single two-cell has not been permitted to carry the argument without supervision.

6.2 Encoding the original labels

Let u_A denote the degree-zero cohomology class of the constant function on S^2_A, and let v_A be its normalized orientation class. Define u_B, v_B similarly. There are isomorphisms

$$F(A) \longrightarrow H^0(S^2_A; \mathbb{R}) \oplus H^2(S^2_A; \mathbb{R}), \qquad e_{a_0} \longmapsto u_A, \qquad e_{a_1} \longmapsto v_A, \tag{6.4}$$

$$F(B) \longrightarrow H^0(S^2_B; \mathbb{R}) \oplus H^2(S^2_B; \mathbb{R}), \qquad e_{b_0} \longmapsto u_B, \qquad e_{b_1} \longmapsto v_B. \tag{6.5}$$

These maps deliberately forget the distinction between degrees zero and two, retaining only even parity. Since the odd cohomology vanishes,

$$[F(U)] = [H^{\mathrm{ev}}(M; \mathbb{R})] - [H^{\mathrm{odd}}(M; \mathbb{R})] \quad \text{in } K_0(\mathbb{R}). \tag{6.6}$$

This is the bridge from the original finite set to geometry. It is explicit rather than canonical under every imaginable change of labels. Canonicity beyond the choices we actually make is unnecessary, and would require another section.

7 Hodge theory and the replacement of classes by solutions

Let $\Omega^j(M)$ be the smooth real differential j-forms on M. The exterior derivative produces the de Rham complex

$$0 \longrightarrow \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \longrightarrow 0. \tag{7.1}$$

The identity $d^2 = 0$ allows its cohomology to be defined. The de Rham theorem identifies that cohomology with $H^*(M; \mathbb{R})$.

The metric and orientation define the Hodge star and the inner product

$$\langle \alpha, \beta \rangle = \int_M \alpha \wedge *\beta. \tag{7.2}$$

Write d^* for the formal adjoint and define the nonnegative Hodge Laplacian

$$\Delta = dd^* + d^*d. \tag{7.3}$$

Integration by parts, with no boundary term, gives

$$\langle \Delta\alpha, \alpha \rangle = \|d\alpha\|^2 + \|d^*\alpha\|^2. \tag{7.4}$$

Thus a form is harmonic precisely when it is closed and coclosed. We write $\mathcal{H}^j(M) = \ker \Delta_j$.

The Hodge theorem asserts that every de Rham cohomology class has a unique harmonic representative on a compact oriented Riemannian manifold [3, Section 8]. Therefore

$$\mathcal{H}^j(M) \cong H^j_{\mathrm{dR}}(M) \cong H^j(M;\mathbb{R}). \tag{7.5}$$

We have exchanged equivalence classes for solutions of an elliptic partial differential equation. This is a substantial improvement if one's objective is to make the problem require partial differential equations.

7.1 The harmonic modes

On a connected compact manifold, a harmonic function is constant: (7.4) gives $df = 0$. On an oriented surface, the Hodge star identifies harmonic zero-forms with harmonic two-forms. On the sphere, $H^1 = 0$ supplies the remaining vanishing.

For $C = A, B$, let $\mathbf{1}_C$ be the function equal to one on S^2_C and zero on the other component. Let

$$\omega_C = \frac{\mathrm{vol}_C}{\int_{S^2_C} \mathrm{vol}_C} \tag{7.6}$$

on its own component, extended by zero elsewhere. The normalization ensures $\int_{S^2_C} \omega_C = 1$. The forms

$$\mathbf{1}_A, \quad \omega_A, \quad \mathbf{1}_B, \quad \omega_B \tag{7.7}$$

span $\mathcal{H}^{\mathrm{ev}}(M)$, and $\mathcal{H}^{\mathrm{odd}}(M) = 0$.

Their linear independence may be checked without a dimension count. If

$$c_0\mathbf{1}_A + c_1\omega_A + c_2\mathbf{1}_B + c_3\omega_B = 0, \tag{7.8}$$

separate degrees, restrict the degree-zero part to each component, and integrate the degree-two part over each component. These operations yield $c_0 = c_1 = c_2 = c_3 = 0$.

One might now conclude the proof by looking at (7.7). We instead ask why infinitely many other eigenfunctions do not affect the answer.

8 The Fredholm problem

Combine the exterior derivative and its adjoint into

$$\mathsf{D} = d + d^*. \tag{8.1}$$

This exchanges even and odd forms and satisfies $\mathsf{D}^2 = \Delta$. Relative to parity,

$$\mathsf{D} = \begin{pmatrix} 0 & \mathsf{D}^- \\ \mathsf{D}^+ & 0 \end{pmatrix}, \qquad \mathsf{D}^- = (\mathsf{D}^+)^*. \tag{8.2}$$

We regard

$$\mathsf{D}^+ : H^1\Omega^{\mathrm{ev}}(M) \longrightarrow L^2\Omega^{\mathrm{odd}}(M) \tag{8.3}$$

as a bounded operator between the indicated Sobolev and Hilbert spaces. Here H^1 denotes Sobolev regularity, not first cohomology; the reuse of the letter is traditional and mildly unhelpful.

Ellipticity on the compact manifold makes D^+ Fredholm. Its kernel and cokernel are finite-dimensional, and elliptic regularity makes their representatives smooth. Using (7.4) and the adjoint,

$$\ker \mathsf{D}^+ = \mathcal{H}^{\mathrm{ev}}(M), \qquad \operatorname{coker} \mathsf{D}^+ \cong \ker \mathsf{D}^- = \mathcal{H}^{\mathrm{odd}}(M). \tag{8.4}$$

Thus

$$\operatorname{ind} \mathsf{D}^+ = \dim \mathcal{H}^{\mathrm{ev}}(M) - \dim \mathcal{H}^{\mathrm{odd}}(M). \tag{8.5}$$

Proposition 8.1 (Index representation of the input)**.** *The integer represented by the original expression equals* $\operatorname{ind} \mathsf{D}^+$*.*

Proof. Apply ρ to (6.6) and use (7.5) and (8.5). The left side is $\dim F(A)+\dim F(B) = |A|+|B|$. □

The proposition is the crucial act of deferral. We have converted an addition problem into an index problem, and the index problem contains the same addition. Its advantage is that the summands are now protected by a substantial analytic vocabulary.

8.1 The zero-dimensional alternative

We could have used the zero-dimensional manifold U itself. Every function on it is harmonic, the exterior derivative is zero, and the index directly records its cardinality. This would avoid spheres, Sobolev spaces, and almost all of the analysis that follows.

We chose a positive-dimensional manifold because otherwise the heat equation would have nothing to do. The final construction still leaves it with nothing to change, but only after cancellation.

9 Heat evolution and cancellation of the positive spectrum

For $t > 0$, the heat operator $e^{-t\Delta}$ is trace class on the L^2 completion of differential forms. Define

$$Z_M(t) = \operatorname{Str}(e^{-t\Delta}) = \operatorname{Tr}(e^{-t\Delta_{\mathrm{ev}}}) - \operatorname{Tr}(e^{-t\Delta_{\mathrm{odd}}}). \tag{9.1}$$

The trace-class property is essential: we are not subtracting two undefined infinite dimensions. Both traces exist before they are subtracted.

Lemma 9.1 (Spectral pairing)**.** *For each eigenvalue $\lambda > 0$, the even and odd eigenspaces of Δ have equal dimensions.*

Proof. The operator D commutes with Δ and reverses parity. On the λ-eigenspace, $\mathsf{D}^2 = \lambda\,\mathrm{id}$. Hence the restriction of D from even to odd is invertible, with inverse $\lambda^{-1}\mathsf{D}$ from odd to even. □

Proposition 9.2 (Heat supertrace formula)**.** *For every $t > 0$,*

$$Z_M(t) = \operatorname{ind} \mathsf{D}^+. \tag{9.2}$$

Proof. The heat trace is the convergent sum of $e^{-t\lambda}$ over eigenvalues counted with multiplicity. By Lemma 9.1, each positive eigenspace contributes equal even and odd terms, which cancel. The zero eigenspace contributes $\dim \mathcal{H}^{\mathrm{ev}}(M) - \dim \mathcal{H}^{\mathrm{odd}}(M)$, since $e^0 = 1$. This is (8.5). □

This is the McKean–Singer heat-supertrace identity in the de Rham case; the classical heat-kernel treatment appears in [4]. We have spelled out the spectral cancellation because it is the exact point where the infinite-dimensional apparatus relinquishes control of the answer.

9.1 Independence of time

Equation (9.2) implies

$$\frac{d}{dt}Z_M(t) = 0 \qquad (t > 0). \tag{9.3}$$

The same conclusion can be viewed through the pairing lemma: every nonconstant exponential term is paired with an equal term of opposite sign. The remaining terms are constant. Heat therefore flows, but the quantity we have chosen to observe refuses to participate.

Taking $t \to \infty$ removes all positive modes from the individual traces as well. The small-time limit $t \downarrow 0$ is more delicate for those traces separately; they diverge. Their supertrace is already constant for every positive time, so no uncontrolled subtraction at $t = 0$ is required.

Remark 9.3 (No regularization of the conclusion). The proof does not assign values to divergent sums. We use ordinary trace-class operators at $t > 0$. The eventual integer is not a zeta-regularized substitute for another integer.

10 An explicit spectral expansion of the same cancellation

The unit round sphere has scalar Laplace eigenvalues

$$\lambda_\ell = \ell(\ell + 1), \qquad \ell = 0, 1, 2, \ldots, \tag{10.1}$$

with multiplicity $2\ell + 1$. These are the spherical-harmonic modes indexed by $m = -\ell, \ldots, \ell$; see [5] for the standard eigenfunction conventions. Define

$$H(t) = \sum_{\ell=1}^{\infty} (2\ell + 1)e^{-t\ell(\ell+1)}. \tag{10.2}$$

For each $t > 0$, this converges, since the exponential decay in ℓ^2 dominates the linear multiplicity.

The degree-zero and degree-two heat traces on one sphere are

$$T_0(t) = 1 + H(t), \qquad T_2(t) = 1 + H(t). \tag{10.3}$$

The second equality follows from the Hodge star. In degree one, there are no harmonic forms. Every positive scalar eigenfunction f generates the exact and coexact eigenforms df and $*df$. On each positive eigenspace these two families span the one-form eigenspace by Hodge decomposition, and are independent. Hence

$$T_1(t) = 2H(t). \tag{10.4}$$

For one sphere,

$$Z_{S^2}(t) = (1 + H(t)) - 2H(t) + (1 + H(t)) = 2. \tag{10.5}$$

For the disconnected union, every trace is a sum over components. Writing the separate components before combining any coefficients,

$$\begin{aligned} Z_M(t) = &\big[(1+H(t)) - 2H(t) + (1+H(t))\big] \\ &+ \big[(1+H(t)) - 2H(t) + (1+H(t))\big]. \end{aligned} \tag{10.6}$$

The only surviving terms are the four constant modes already listed in (7.7). The series has confirmed their survival by introducing every positive angular frequency and then canceling it.

10.1 What the explicit expansion establishes

Formula (10.6) is an analytic certificate of the same ordinary arithmetic, not an arithmetic-free route to it. In particular, simplifying coefficients in a spectral formula is still arithmetic. The force of the calculation is that its modes, signs, and multiplicities agree with the finite-set model, rather than that it has replaced addition with an operation of unknown origin.

The raw trace of $e^{-t\Delta}$ on all forms would retain the positive modes and depend on time. It is the *supertrace*, with the odd-degree minus sign, that extracts the desired invariant. Removing that sign would not simplify the proof. It would change the quantity being computed.

11 Curvature as an additional accounting system

The Gauss–Bonnet theorem for a closed oriented surface states

$$\chi(M) = \frac{1}{2\pi}\int_M K\, dA, \tag{11.1}$$

where K is Gaussian curvature; see [3, Section 7]. On a disjoint union, both sides are sums of the corresponding component quantities.

For a round sphere of radius $R > 0$, $K = R^{-2}$ and the area element in spherical coordinates is $R^2 \sin\theta\, d\theta\, d\phi$. Thus

$$\begin{aligned} \frac{1}{2\pi}\int_{S^2_R} K\, dA &= \frac{1}{2\pi}\int_0^{2\pi}\int_0^{\pi} R^{-2} R^2 \sin\theta\, d\theta\, d\phi \\ &= \frac{1}{2\pi}\,(2\pi)\,[-\cos\theta]_0^{\pi} \\ &= 2. \end{aligned} \tag{11.2}$$

The radius cancels. The factors of π cancel. The trigonometric function is evaluated at the endpoints. The original small integer remains.

For $M = S^2_A \sqcup S^2_B$, this gives

$$\chi(M) = \frac{1}{2\pi}\left(\int_{S^2_A} K\, dA + \int_{S^2_B} K\, dA\right). \tag{11.3}$$

Combining Gauss–Bonnet with Hodge theory and Proposition 9.2 identifies this curvature expression with $Z_M(t)$ and $\operatorname{ind} \mathsf{D}^+$. There are now three languages in which the same modes can be counted, and none of them changes their number.

11.1 Independent radii

The two components need not have equal radii. Let their radii be R_A and R_B. Each curvature integral is still given by (11.2). Their positive Laplace spectra are generally different, since eigenvalues scale as R_C^{-2}. Nevertheless, positive-mode cancellation occurs *within each component*, between parities. No matching of frequencies across the components is required.

This observation eliminates a possible misunderstanding: the two copies of two do not need to be the same physical size. Their role is cohomological. Enlarging one sphere changes its geometry without providing an additional contribution to the answer.

12 Robustness, functoriality, and changes that cannot help

Proposition 12.1 (Metric independence)**.** *For every smooth Riemannian metric on the fixed oriented manifold M, the heat supertrace of its Hodge Laplacian equals the same Euler characteristic.*

Proof. Hodge theory identifies the harmonic spaces with the fixed de Rham cohomology groups. Spectral pairing cancels every positive eigenspace for the chosen metric. Hence the supertrace depends only on the alternating cohomology dimensions. □

The conclusion survives a lumpy sphere. The explicit spherical harmonics of Section 10 do not survive in that form, but the index does. The special metric was used to produce a particularly ornate formula, not to stabilize an otherwise uncertain addition.

12.1 Relabeling

Permuting a_0, a_1 or b_0, b_1 permutes the corresponding free basis. After forgetting integer degree and retaining even parity, this merely changes the chosen identification with harmonic modes. It does not change the dimension or index. Exchanging the two input sets exchanges the two components of M.

The construction therefore respects the commutativity of the original addition. We do not need to prove commutativity of all addition to evaluate this particular symmetric expression, but it is reassuring that a geometric implementation does not accidentally remember which two was on the left.

12.2 Stabilization

Adding $E_r^\bullet$ from Section 5 leaves the Euler class unchanged. In a finite-dimensional Hilbert realization, the differential on each paired coordinate is the identity, so its Laplacian is the identity in both adjacent degrees. Its heat supertrace is

$$-re^{-t} + re^{-t} = 0. \tag{12.1}$$

The same cancellation is visible algebraically, homologically, and spectrally. Nothing has acquired three mutually consistent descriptions.

12.3 Limits of the robustness statement

We have not shown that arbitrary changes to the manifold preserve the answer. Replacing a sphere by a torus changes the cohomology, and adding a boundary introduces analytic choices that are absent here. Robustness concerns the specified representation of the original finite sets. It does not license replacing the problem by a different one and retaining its conclusion.

13 Descent to the natural numbers

We now identify the finite obstruction that survives the preceding analysis. Define

$$\Phi : F(Q) \longrightarrow \mathcal{H}^{\mathrm{ev}}(M) \tag{13.1}$$

on the free basis by

$$e_{q_0} \mapsto \mathbf{1}_A, \qquad e_{q_1} \mapsto \omega_A, \qquad e_{q_2} \mapsto \mathbf{1}_B, \qquad e_{q_3} \mapsto \omega_B. \tag{13.2}$$

The spanning and independence argument in Section 7 proves that Φ is an isomorphism. Since $\mathcal{H}^{\mathrm{odd}}(M) = 0$, this gives

$$[F(A)] + [F(B)] = [F(U)] = [\mathcal{H}^{\mathrm{ev}}(M)] - [\mathcal{H}^{\mathrm{odd}}(M)] = [F(Q)]. \tag{13.3}$$

Proof of Theorem 1.1. Apply the rank isomorphism $\rho : K_0(\mathbb{R}) \to \mathbb{Z}$ to (13.3). The two input sets are each identified with the standard ordinal $\underline{2}$, and Q is identified with $\underline{4}$. Therefore the resulting integer equality is

$$2 + 2 = 4. \tag{13.4}$$

The canonical embedding $\mathbb{N} \hookrightarrow \mathbb{Z}$ preserves addition and is injective, so the equality also holds in $\mathbb{N}$. □

The analytic portion supplies the compatible identity

$$\boxed{2 + 2 = \operatorname{ind} \mathsf{D}^{+} = \operatorname{Str}(e^{-t\Delta_M}) = \chi(M) = \frac{1}{2\pi} \int_M K \, dA = 4, \quad t > 0.} \tag{13.5}$$

Each equality has been assigned a meaning, domain, and justification. The heat equation has delivered the answer in the same condition in which it was handed to the category of finite sets.

Corollary 13.1 (Persistence under observation)**.** *The conclusion remains true if the intermediate constructions are not inspected.*

Proof. Inspection was not a hypothesis of Theorem 1.1. □

14 Discussion of efficiency and mathematical debt

The central difficulty was never determining the answer. It was maintaining a chain of correctly typed constructions long enough to make the answer look as though it had survived an investigation. The proof has required no conjectures, numerical tolerances, or unverified computations. It has required a disproportionate amount of established theory.

14.1 Dependency audit

Table 1 records the principal additions to the problem. "Removable" means removable from a proof of the original theorem, not removable from every argument in the section that uses it.

Structure	Assigned responsibility	Removable
Tags	Prevent accidental identification	Yes
Coproduct universal property	Re-express the tagged union	Yes
Free vector spaces	Replace objects by coordinates	Yes
Grothendieck group	Permit cancellation	Yes
Contractible complex	Create terms to cancel	Yes
Disconnected surface	Introduce geometry	Yes
Hodge theorem	Choose harmonic representatives	Yes
Fredholm operator	Package an alternating dimension	Yes
Heat semigroup	Suppress already paired modes	Yes
Gauss–Bonnet	Obtain the same number from curvature	Yes
Spectral sequence	Verify a filtration with one column	Yes
Recursive addition	Evaluate the original expression	No*

Table 1: A record of mathematical expenditure. The last entry is essential to the elementary proof in Appendix A; equivalent definitions of addition may replace it.

The table should not be interpreted as criticism of the theories listed. Each has important applications. The present application is not offered as evidence of that importance.

14.2 Why the proof cannot be maximally inefficient

Suppose one has completed the manuscript. Insert a new lemma stating that the identity map on $F(U)$ is an isomorphism, prove it by writing down its inverse, and renumber the subsequent results. The new proof is longer. The process can be repeated. Alternatively, append a fresh contractible summand and verify its contraction coordinate by coordinate.

Thus no finite manuscript deserves the literal title "the longest proof." Our construction provides a finite witness to poor judgment, not an extremum in the partially ordered set of poor judgments. We stop when the overhead has become unmistakable, which is an editorial criterion rather than a mathematical bound.

14.3 Computational complexity

For this fixed input, both the elementary proof and the entire construction can be described as constant-size computations once all conventions are fixed. Big-O notation is therefore an unsatisfactory measure of the damage. The hidden constant contains the Hodge theorem.

No floating-point experiment is needed. A numerical approximation to a heat trace would introduce truncation and rounding errors into a problem that began with exact finite data. Appendix E instead records finite structural checks. They verify the bookkeeping without pretending to verify every theorem of elliptic analysis.

15 Conclusion

We have shown that the additive composition of two and two is four. The result is stable under changes of metric, relabeling, and the addition of contractible complexes. It also agrees with counting.

The extension to three plus three is left for a separate paper. A naive adaptation suggests three spheres; whether this is an appropriate use of the reader's remaining time has not been established.

A Foundational normalization of the numerals

For completeness, we give the proof that the main text has been systematically avoiding. Define the numerals by

$$1 = S(0), \qquad 2 = S(S(0)), \qquad 3 = S(S(S(0))), \qquad 4 = S(S(S(S(0)))). \tag{A.1}$$

Addition is specified recursively by

$$a + 0 = a, \qquad a + S(b) = S(a + b). \tag{A.2}$$

Substituting these definitions yields

$$\begin{aligned} 2 + 2 &= S(S(0)) + S(S(0)) && \text{(A.3)}\\ &= S\big(S(S(0)) + S(0)\big) && \text{(A.4)}\\ &= S\big(S\big(S(S(0)) + 0\big)\big) && \text{(A.5)}\\ &= S\big(S\big(S(S(0))\big)\big) && \text{(A.6)}\\ &= 4. && \text{(A.7)} \end{aligned}$$

The first and last steps expand or recognize numerals; the middle steps apply (A.2). No commutativity, associativity, induction over an unbounded family, or geometric construction is needed for this particular calculation.

The recursion theorem that defines addition belongs to the background foundations. Once the two recursion equations are available, the displayed chain is a direct equational derivation. This is the appropriate answer if the question is genuinely about proving the identity from the defining equations, rather than about how much mathematics can be made to attend.

A.1 Set-theoretic realization

In the von Neumann realization,

$$0 = \varnothing, \qquad S(n) = n \cup \{n\}. \tag{A.8}$$

Successive applications give

$$\begin{aligned} 1 &= \{\varnothing\}, && \text{(A.9)}\\ 2 &= \{\varnothing, \{\varnothing\}\}, && \text{(A.10)}\\ 3 &= \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}, && \text{(A.11)}\\ 4 &= \{0, 1, 2, 3\}. && \text{(A.12)} \end{aligned}$$

We have abbreviated the last line because expanding all of its braces would produce length without a corresponding increase in spectral content. There must be limits.

A.2 Relation to the main proof

This appendix supplies the small formal derivation underlying the familiar arithmetic used elsewhere. The main proof supplies compatible realizations in other mathematical settings. Reading them in this order is logically harmless and pedagogically questionable. Reading this appendix first would have been more efficient, which explains its position.

B The terminal matching table

The bijection $U \to Q$ is given by the first and last columns below. The intermediate columns record the structures traversed in the paper.

Tagged element	Free basis	Harmonic representative	Standard label
(a_0, L)	$e_{(a_0,L)}$	$\mathbf{1}_A$	q_0
(a_1, L)	$e_{(a_1,L)}$	ω_A	q_1
(b_0, R)	$e_{(b_0,R)}$	$\mathbf{1}_B$	q_2
(b_1, R)	$e_{(b_1,R)}$	ω_B	q_3

Every member of the domain occurs once. Every member of the codomain occurs once. Reversing the rows defines an inverse. This establishes injectivity and surjectivity without requiring a cardinality formula.

Lemma B.1 (Commutativity of the completed itinerary)**.** *Following a row through the intermediate identifications gives the same terminal label as applying the direct bijection.*

Proof. The maps were defined on the displayed generators by those rows. Equality on a basis implies equality of the induced linear maps. The remaining finite-set statement follows by inspection of the domain labels. □

The direct bijection is already a proof of the cardinal identity. The rest of the paper establishes that this bijection can be reached after an extensive and fully reversible loss of perspective.

C A spectral sequence with nowhere to go

Let $C^\bullet = \widetilde{F}(U)^\bullet$ be the stabilized complex from (5.3). Give it the decreasing filtration

$$F^p C^\bullet = \begin{cases} C^\bullet, & p \le 0, \\ 0, & p \ge 1. \end{cases} \tag{C.1}$$

The only nonzero associated-graded column is $p = 0$. The spectral sequence of this filtered complex has

$$E_0^{0,q} = C^q, \qquad E_1^{0,q} = H^q(C^\bullet), \tag{C.2}$$

and $E_1^{p,q} = 0$ for $p \neq 0$. Since E_r has no cohomology, only $E_1^{0,0} = F(U)$ remains.

For $s \geq 1$, the differential has bidegree $(s, 1-s)$:

$$d_s : E_s^{p,q} \longrightarrow E_s^{p+s,q-s+1}. \tag{C.3}$$

Every possible differential from the surviving term has zero target, and every possible incoming differential has zero source. Therefore

$$E_1 = E_2 = E_3 = \cdots = E_\infty. \tag{C.4}$$

The filtration is finite on each term and the complex is bounded, so the spectral sequence converges to the associated graded of $H^*(C^\bullet)$. There is only one filtration piece. Accordingly, there is no extension problem.

Proposition C.1 (Early degeneration)**.** *The spectral sequence recovers $F(U)$ and supplies no new numerical information.*

Proof. All pages after the first are forced by their empty surroundings. □

For clarity, the auxiliary complex is called $E_r^\bullet$, whereas the pages are called $E_s^{p,q}$. The potential collision of letters is avoided by the indices. A more considerate author would have changed one of the letters.

D A determinant and an unused meromorphic continuation

The finite-dimensional harmonic space offers another way to encode the same integer. Define

$$P(z) = \det\bigl(\mathrm{id} + z\,\mathrm{id} \mid \mathcal{H}^{\mathrm{ev}}(M)\bigr). \tag{D.1}$$

Using the ordered basis (7.7), the matrix is diagonal with entries $1+z$ in each of its four positions. Thus

$$P(z) = (1+z)^4, \tag{D.2}$$

and its logarithmic derivative at zero is

$$\frac{d}{dz}\log P(z)\Big|_{z=0} = 4. \tag{D.3}$$

Equivalently, the derivative equals the trace of the identity on the harmonic space. This calculation does not make the answer more exact. It demonstrates that the answer can be extracted from a polynomial after being inserted into its degree.

D.1 The super zeta function of the discarded modes

Complexify the form spaces solely to use standard complex spectral notation. For $\Re(s) > 1$, define the positive-spectrum super zeta function on the surface by

$$\zeta_{\mathrm{sup}}(s) = \sum_{\lambda>0} \bigl(m_{\mathrm{ev}}(\lambda) - m_{\mathrm{odd}}(\lambda)\bigr)\lambda^{-s}. \tag{D.4}$$

The individual degreewise zeta sums converge in this half-plane. By Lemma 9.1, every coefficient in the displayed sum is zero. Hence

$$\zeta_{\mathrm{sup}}(s) = 0 \qquad (\Re(s) > 1). \tag{D.5}$$

The zero function extends to the entire complex plane. No general meromorphic-continuation theorem is needed to extend a function already known to be zero.

This function does *not* return four. Its definition deliberately omits the zero modes, and those are precisely the modes that carry the answer. Any attempt to recover the theorem from this zeta function alone would have discarded the relevant data before continuing it analytically.

We include the calculation because the words "meromorphic continuation" might otherwise be missed, and because it identifies one more sophisticated procedure that is incapable of rescuing information it was instructed to remove.

E Finite verification and an ablation study

The structural checks accompanying this paper concern the explicit bijections, contractions, and finite matrices. They do not test the Hodge theorem by sampling manifolds.

E.1 Checks that can actually fail

The tagged union must have no repeated members. The map in Appendix B must be total, injective, and surjective. The proposed inverse must compose to the identity on both sides. For the complex $E_r^\bullet$, matrix multiplication must give $d^2 = 0$ and $dh + hd = \mathrm{id}$. The parity-folded stabilization must have zero superdimension. These checks would detect a missing label, an incorrectly signed degree, or a contraction placed in the wrong direction.

The scalar spectral terms can also be checked with exact formal bookkeeping. For each positive angular degree ℓ, let $m = 2\ell + 1$. The contributions in degrees zero, one, and two are $m, -2m, m$. Their sum is zero. For the zero modes, the corresponding multiplicities are $1, 0, 1$ on each component. This check addresses signs and multiplicities; it is not a substitute for the analytic fact that the heat operators are trace class.

E.2 Ablation protocol

Remove the determinant appendix. The theorem survives. Remove the spectral sequence. The theorem survives. Remove the curvature calculation, the spherical-harmonic expansion, and the heat kernel. The theorem survives by the explicit harmonic basis. Remove the harmonic basis, the manifold, and the Grothendieck group. The theorem survives by the matching table. Remove the table. The recursive derivation still proves the theorem.

This is a theoretical dependency analysis, not an empirical performance study. The observed robustness to deletion should not be confused with evidence that the deleted material was necessary.

E.3 Reproducibility statement

The LaTeX source accompanies the paper. The finite check script uses exact integer arithmetic and the Python standard library. Its result can be reproduced without a GPU, a random seed, or an account on a platform. The full analytic argument remains a proof to be read, not a software result to be trusted because a process exited successfully.

F Illustrative referee objections and author responses

The following exchange is fictional and forms part of the satire. It is not a record of peer review.

Objection 1: The result is obvious

Referee. The conclusion can be obtained by counting four objects.

Response. We thank the referee for describing the terminal stage of our method. The revised manuscript now explains how those objects may first be replaced by harmonic representatives.

Objection 2: The proof uses ordinary arithmetic

Referee. The advanced theorems already presuppose the arithmetic you purport to establish.

Response. Correct. Section 1 distinguishes an internal derivation from an independent foundation, and Appendix A provides the direct recursive proof. The paper's purpose is an unreasonable route to a known conclusion, not a claim to have made arithmetic depend historically or logically on the heat equation.

Objection 3: The derived category can be deleted

Referee. I cannot identify any use of derived stabilization that is essential to the result.

Response. We have added an ablation study confirming the referee's observation. Deleting the section improves the proof but reduces the manuscript's contribution to its stated objective.

Objection 4: The spectral sequence degenerates immediately

Referee. There is only one nonzero entry. The spectral sequence has no work to perform.

Response. Its early degeneration is now stated as a proposition. We believe this addresses the concern at the appropriate level of formality.

Objection 5: Why not use a zero-dimensional manifold?

Referee. The finite set itself is a compact zero-dimensional manifold and gives the index immediately.

Response. This alternative is acknowledged in Section 8. It was excluded because it would avoid the positive spectrum and leave the heat kernel insufficiently occupied.

Objection 6: The paper is far too long

Referee. I recommend reducing the manuscript to one line.

Response. We attempted this revision and found that it removed every section in which the present work differs from existing treatments.

Objection 7: What has been learned?

Referee. The numerical conclusion appears unchanged from the beginning.

Response. The conclusion is unchanged. The reader is not.

Acknowledgments

The author thanks the reader for the compact interval of their life on which this argument was supported. The standard theories cited below bear no responsibility for the use to which they have been put.

Availability and declarations

Data availability. All input objects appear in Section 2. No additional objects are available on request.

Limitations. The method has not been optimized. The conclusion has.

Competing interests. The principal competing interest is the reader's time.

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